

Auditing Predictive Information in SEC 10-K Text: Evidence from a 50-Firm U.S. Large-Cap Pilot

Tom Lin^a

*^aDepartment of Mathematical Sciences, Beijing Normal-Hong Kong Baptist University
(BNBU), Zhuhai, China*

Abstract

SEC 10-K filings contain firm-level disclosures about operations, risk, litigation, and management expectations, but textual prediction studies are difficult to interpret when timing, model selection, and specification search are not fully audited. This study examines whether 10-K textual features contain out-of-sample information about future realized volatility and abnormal-return targets in a fixed 50-firm U.S. large-cap pilot covering fiscal years 2016–2025. The pipeline links SEC EDGAR filings to event-time labels, parses full filings and economically interpretable sections, constructs Loughran-McDonald dictionary features and train-window-only TF-IDF/SVD features, and evaluates historical mean, industry mean, Ridge, and XGBoost models under expanding rolling splits with validation-only model selection. The pre-specified primary prediction specification, Ridge on `realized_volatility_1_20`, achieves an ALL_SPLITS Rank IC of 0.2395 with a raw p-value of 0.00067. The strongest observed ranking result is a TF-IDF/SVD-only Ridge robustness comparison, with Rank IC of 0.3668 and industry-neutral Rank IC of 0.3416. Bootstrap diagnostics make the interpretation cautious: the primary raw Rank IC is positive, but the split-level bootstrap interval includes zero, and all primary industry-neutral bootstrap intervals include zero. Evidence for abnormal-return targets is weaker, and the pre-specified monthly sector-neutral portfolio diagnostic has Sharpe ratio -0.8539 with p-value 0.1147. The contribution is an auditable, leakage-aware financial NLP workflow that separates realized-volatility prediction evidence from portfolio, multiple-testing, and data-boundary limitations.

Keywords: financial NLP, disclosure text, realized volatility, out-of-sample prediction, leakage control, multiple testing, SEC filings

1. Introduction

Annual reports are among the most widely observed disclosure documents in U.S. capital markets. A Form 10-K reports accounting outcomes, but it also describes business models, operating risks, legal exposure, liquidity constraints, and management’s interpretation of performance. Because these disclosures are available before future return and volatility realizations, they provide a natural setting for testing whether firm-level text contains predictive information. The empirical difficulty is that textual prediction can be overstated when document timing, label windows, model selection, and specification search are not audited. A positive text signal may reflect genuine information, but it may also reflect leakage, post hoc tuning, or an unreported search over many alternatives.

This working paper studies that problem through a bounded 50-firm pilot. The current run covers fiscal years 2016–2025 and links 500 SEC 10-K filings to 1,500 event-time labels, 520,465 text feature records, and 8,133 rolling out-of-sample predictions. The pipeline parses the full filing and four sections: Item 1 Business, Item 1A Risk Factors, Item 3 Legal Proceedings, and Item 7 Management’s Discussion and Analysis. It then constructs Loughran-McDonald dictionary-style tone features and TF-IDF/SVD features whose vocabularies are fit only within each training window. Historical mean, industry mean, Ridge, XGBoost, and Ridge feature-ablation variants are evaluated under expanding rolling train-validation-test splits. The v4 evaluation layer also adds industry-neutral Rank IC, bootstrap confidence intervals, parser-review diagnostics, and a larger multiple-testing registry.

The study asks four related questions. Do 10-K text features contain out-of-sample ranking information for future 20-day realized volatility? Do the same features predict the abnormal-return targets `CAR_1_5` and `CAR_1_20`? Do prediction scores produce informative portfolio diagnostics after monthly common rebalancing, transaction costs, and sector-neutral construction rules? Finally, how much of the evidence remains after pre-specified specification roles and multiple-testing disclosure? These questions separate prediction from trading interpretation. A model may rank future volatility without producing a cost-adjusted long-short equity strategy.

The emphasis on realized volatility is substantive rather than opportunistic. A 10-K describes risk factors, litigation exposure, liquidity conditions, business complexity, and management’s assessment of operating conditions. These disclosures are closely related to future uncertainty and therefore provide a

plausible channel for volatility prediction. Abnormal returns are harder to interpret from the same text because prices may already reflect part of the disclosure, expected returns combine risk compensation with mispricing, and short-window returns contain substantial market and firm-specific noise. The paper therefore treats future realized volatility as the main empirical target and treats abnormal-return and portfolio results as diagnostics that test the economic reach of the same signal.

The primary result is concentrated in volatility prediction. The pre-specified primary prediction specification uses combined-text Ridge to predict the 20-day realized-volatility target and evaluates ALL_SPLITS Rank IC on out-of-sample test predictions. This specification produces a Rank IC of 0.2395 with a raw p-value of 0.00067. The strongest observed model-comparison result is not XGBoost in v4, but TF-IDF/SVD-only Ridge, with Rank IC of 0.3668, Newey-West t-statistic of 5.4055, RMSE of 0.00992, and industry-neutral Rank IC of 0.3416. This result is informative, but it remains a robustness comparison interpreted as exploratory relative to the pre-specified Ridge combined-text primary claim.

Evidence for the abnormal-return targets `CAR_1_5` and `CAR_1_20` is more limited, and the pre-specified monthly sector-neutral portfolio diagnostic does not establish formal trading alpha. The appropriate conclusion is therefore specific: the current public package supports audit-backed out-of-sample prediction evidence for future realized volatility, not a formal asset-pricing result.

The paper contributes a reproducible research design for financial text factors. It records document manifests, event-time alignment, parsed sections, labels, rolling splits, feature manifests, train-window vocabulary hashes, model manifests, predictions, evaluation metrics, portfolio diagnostics, tested specifications, multiple-testing adjustments, and audit reports. The audit layer reports 100 percent eligible out-of-sample coverage, 98.55 percent model-expected prediction coverage, 100 percent primary prediction and portfolio coverage, zero audit failures, and two audit warnings. The data-boundary limitation is treated explicitly in Section 3 rather than converted into an asset-pricing claim. The remainder of the paper positions the study, describes the data and methods, reports prediction and portfolio diagnostics, and states the limits of the current applied-grade evidence.

2. Related Literature and Positioning

This paper relates to five strands of work: textual disclosure and the firm information environment, financial NLP feature construction, out-of-sample prediction protocols, the distinction between prediction and asset-pricing claims, and reproducible empirical design. The review is selective because the paper’s empirical contribution is not a new theory of disclosure. The contribution is a controlled research design for asking whether information embedded in 10-K text survives timing, feature-construction, model-selection, and multiple-testing controls.

2.1. Textual disclosure and firm information environment

SEC filings are a natural source for studying firm information because they combine mandatory disclosure, standardized filing timestamps, and economically interpretable narrative sections. Work on corporate filing text has shown that disclosure language contains information beyond accounting numbers, including tone, readability, forward-looking discussion, financial constraints, political risk, and changes in risk-related language (Loughran and McDonald, 2011; Li, 2008, 2010; Buehlmaier and Whited, 2018; Hassan et al., 2019; Cohen et al., 2020). Risk-factor and MD&A disclosures are especially relevant for this paper because they describe uncertainty, litigation, liquidity, competitive conditions, and management’s assessment of operating performance. These components plausibly affect the distribution of future outcomes even when they do not translate cleanly into signed abnormal returns.

The volatility emphasis follows this disclosure logic. Text describing risks and uncertainty may be more directly connected to future return variability than to abnormal returns. Abnormal returns depend on market expectations, risk compensation, and the degree to which public information is already capitalized into prices. By contrast, future realized volatility can respond to uncertainty about the range of firm outcomes. This distinction is consistent with studies that use disclosure text to measure risk and uncertainty rather than only directional sentiment (Kogan et al., 2009; Campbell et al., 2014; Manela and Moreira, 2017).

2.2. Financial NLP feature construction

Financial text studies differ sharply in how they represent language. Dictionary approaches use pre-defined term lists, most prominently finance-specific

108 dictionaries designed to avoid misclassifying ordinary financial language as
109 negative sentiment (Loughran and McDonald, 2011). These features are
110 transparent and easy to audit, but they compress language into a small
111 set of predetermined categories. Statistical text representations, including
112 TF-IDF, topic models, embeddings, and more recent language-model features,
113 can capture richer variation in documents but introduce additional design
114 choices about vocabulary fitting, dimensionality reduction, and model selec-
115 tion (Gentzkow et al., 2019; Hoberg and Phillips, 2016; Lopez-Lira and Tang,
116 2023).

117 The current paper is positioned between these approaches. It keeps
118 dictionary features because they make term sources, feature names, and
119 economic interpretation visible. It also uses TF-IDF/SVD features because
120 10-K sections may contain predictive variation that is not captured by a fixed
121 dictionary. The key methodological point is not that either representation
122 is novel. It is that train-window-only fitting makes the representation itself
123 part of the out-of-sample protocol.

124 *2.3. Out-of-sample prediction and leakage control*

125 Text prediction is vulnerable to subtle forms of look-ahead bias. Full-
126 sample vocabularies, post-event labels, delayed filing timestamps, and model
127 choices made after seeing test performance can all inflate apparent predictabil-
128 ity. The concern is stronger in high-dimensional text settings because many
129 targets, sections, representations, and models can be searched before the final
130 result is reported. Economic text-as-data work therefore requires not only a
131 prediction model but also a documented timing convention and a defensible
132 training protocol (Gentzkow et al., 2019; White, 2000).

133 This paper treats leakage control as an empirical design object. The
134 local pipeline records filing acceptance times, event-time labels, rolling train-
135 validation-test splits, train-window vocabulary hashes, model manifests, tun-
136 ing logs, tested specifications, and audit reports. Model selection is based
137 on validation windows rather than test performance. This design directly
138 addresses the concern that a textual factor may look predictive because the
139 feature pipeline or specification search has used information that would not
140 have been available at the prediction date.

141 *2.4. From prediction to asset-pricing claims*

142 Prediction evidence is not equivalent to an asset-pricing result. A model
143 can rank future realized volatility or even produce statistically nonzero return

forecasts without establishing a cost-adjusted tradable strategy. Portfolio evidence requires additional assumptions about turnover, transaction costs, investability, weighting, rebalancing time, sector exposure, and the data universe. It also requires caution about multiple testing, because many candidate signals can appear successful by chance when researchers search across specifications (White, 2000; Harvey et al., 2016; Bailey et al., 2014; Benjamini and Hochberg, 1995).

The paper therefore separates three claims. The first is a prediction claim: whether text scores rank future outcomes out of sample. The second is a portfolio-diagnostic claim: whether the same scores remain informative after a specified trading rule. The third is a formal asset-pricing claim, which would require a broader research-grade data design. The current run supports the first claim for future realized volatility, provides limited evidence for the abnormal-return targets CAR_1_5 and CAR_1_20, and does not support a formal trading-alpha claim. This separation is consistent with the empirical asset-pricing distinction between predictive scoring, abnormal-return adjustment, and factor-model interpretation (Fama and French, 1993; Carhart, 1997; Fama and French, 2015; Gu et al., 2020).

2.5. Positioning of this paper

The paper’s contribution is the audit trail connecting disclosure text to empirical interpretation. Compared with studies that report a final text metric or a single model comparison, this design records the construction path: document manifests, filing-time alignment, parsed sections, labels, rolling splits, feature manifests, train-window vocabulary hashes, validation-only model selection, predictions, evaluation metrics, portfolio diagnostics, specification roles, multiple-testing adjustments, and audit outcomes. The run-level specification registry identifies the primary prediction and portfolio rules before interpretation; the language in this paper describes them as pre-specified rather than formally registered.

The scope remains deliberately bounded. The current evidence comes from a fixed 50-company applied panel and a mixed FMP/Yahoo public-source market-data stack. It is not a CRSP/WRDS-equivalent survivorship-free asset-pricing study. Within that boundary, the paper contributes a reproducible workflow for testing financial text factors and an empirical finding that 10-K text contains positive out-of-sample ranking information for future 20-day realized volatility.

Table 1: Sample construction summary for the 50-company 10-K text sample

Component	Local project record
Universe	50 unique entities, 50 unique tickers, 11 GICS-style sectors, selection date January 1, 2016.
Filings	500 SEC EDGAR Form 10-K filings, covering fiscal years 2016–2025 with 50 filings in each fiscal year.
Parsed text	Full filing plus Item 1 Business, Item 1A Risk Factors, Item 3 Legal Proceedings, and Item 7 MD&A; 2,000 parsed section records; zero missing raw filings.
Market data	133,518 price-panel rows from December 1, 2015 to April 30, 2026 across 51 tickers, including SPY as benchmark; FMP adjusted-close data with Yahoo fallback allowed.
Labels	1,500 labels: 500 observations each for <code>realized_volatility_1_20</code> , <code>CAR_1_5</code> , and <code>CAR_1_20</code> .
Feature records	520,465 feature records and 40 feature-manifest entries, combining dictionary tone features with train-window-only TF-IDF/SVD features.

180 3. Data and Universe Construction

181 The empirical sample is defined by the current 50-company public FMP
182 alpha run. The universe is a fixed 50-company U.S. large-cap sector-seed
183 panel with selection date January 1, 2016. The fixed panel makes the pilot
184 auditable and reproducible: every company, filing, label, and prediction
185 can be traced through the local artifacts. The cost of that design is also
186 explicit. The panel is classified as *applied* rather than research-grade because
187 all 50 market-cap-at-selection entries are seed-rank estimates rather than
188 CRSP/WRDS-style historical market capitalizations, and the universe does
189 not claim survivorship-free construction.

190 Figure 1 reports the sector composition used by the local universe file.
191 The sector seed is not equal-weighted across sectors, but it is broader than a
192 technology-only or financial-only sample. Financials account for the largest
193 share with eight firms, followed by Information Technology and Health Care
194 with six firms each. Real Estate is the smallest sector with two firms. This
195 sector spread is useful for a pilot because it exposes the parsing and feature

Table 2: Audit and data-boundary summary

Boundary item	Interpretation for this paper
Run provenance	Run status reported ; package version 0.16.0; git commit 753240b; run-level audit status warn .
Universe status	Applied fixed-company panel; useful for an auditable pilot, but not a survivorship-free research-grade universe.
Market data	Mixed FMP/Yahoo public-source stack; suitable for applied prediction diagnostics, but not CRSP/WRDS-equivalent portfolio evidence.
Parser quality gate	Very short Item 1A or Item 7 extractions are excluded from section-level features until parser boundaries are manually reviewed.
Coverage interpretation	Raw label coverage is 0.496 because the denominator includes labels outside the configured validation/test windows; eligible out-of-sample and model-expected prediction coverage are both 1.000.
Formal result boundary	Zero audit failures and two audit warnings support an applied-grade prediction claim, not a formal trading-alpha or asset-pricing claim.

196 pipeline to different disclosure styles, but it does not remove the applied-panel
197 limitation.

198 The filing layer is constructed from SEC EDGAR 10-K documents. Each
199 document-manifest row records the entity identifier, ticker, CIK, fiscal year,
200 source identifier, retrieval time, SEC acceptance time, event time, resolved
201 event date, market-calendar status, hash, license note, and parser version.
202 This design makes the filing date part of the empirical object rather than a
203 background assumption. It is included to prevent filing-date look-ahead bias:
204 labels are aligned to information that was available at prediction time, not to
205 the fiscal year alone.

206 The parser produces one full-document representation and four section-
207 level representations. The project-level parsing report records 500 total
208 documents, zero missing raw filings, and 2,000 parsed section records. The
209 section-length report records 1,840 sections marked **ok**, 16 sections with a
210 warning for fewer than 500 words, and 144 sections marked for manual review
211 below 100 words. The same report records 144 section-level feature exclusions

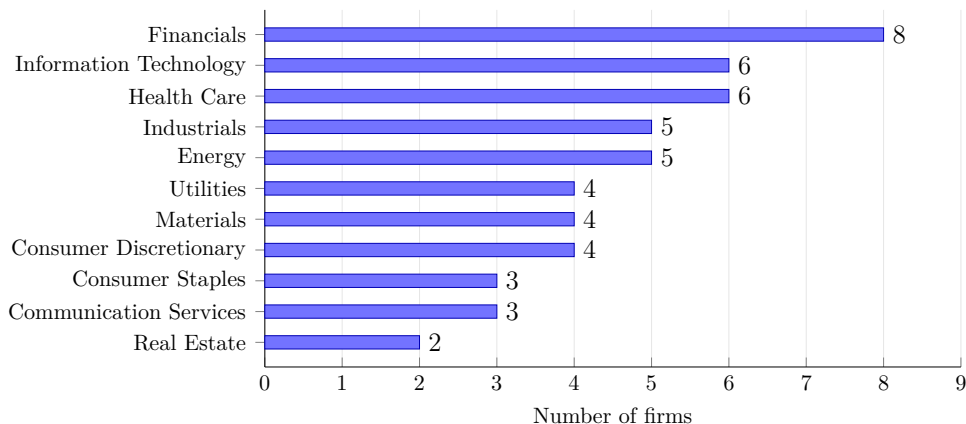


Figure 1: Sector distribution of the fixed 50-company applied universe. The spread across 11 sectors reduces dependence on one disclosure style, but it does not remove the fixed-panel and non-survivorship-free data-boundary limitation.

212 under the policy that very short Item 1A or Item 7 extractions are not used
 213 for section-level features until parser boundaries are manually reviewed. This
 214 rule is conservative: it retains document-level coverage while preventing parser
 215 boundary errors from contaminating section-level text features.

216 The market-data layer uses adjusted prices and returns from the pub-
 217 lic FMP/Yahoo stack recorded in the configuration. The run-level price
 218 file contains 133,518 rows from December 1, 2015 through April 30, 2026,
 219 which allows post-filing labels for filings near the end of FY2025. Labels
 220 are event-time outcomes beginning after the filing event. The volatility tar-
 221 get, `realized_volatility_1_20`, is computed from post-event log returns
 222 without a market benchmark. The two abnormal-return targets, `CAR_1_5`
 223 and `CAR_1_20`, use SPY as the benchmark and close-adjusted log returns.
 224 The mixed source is not a pipeline failure, but it is a data-boundary limita-
 225 tion for market-data-dependent portfolio claims. The delisting application
 226 report records 1,500 labels, zero delisting returns applied, zero labels missing
 227 delisting returns, and status `not_applicable`; this is another applied-data
 228 boundary rather than a CRSP-equivalent delisting treatment.

229 The split layer uses expanding rolling year splits. The first split trains on
 230 2016–2020 filings, validates on 2021, and tests on 2022; the final split trains
 231 on 2016–2023, validates on 2024, and tests on 2025. The split assignment
 232 file contains 4,574 assignment records across the four split definitions. The
 233 audit report records 46 split-leakage records purged by the 20-day embargo

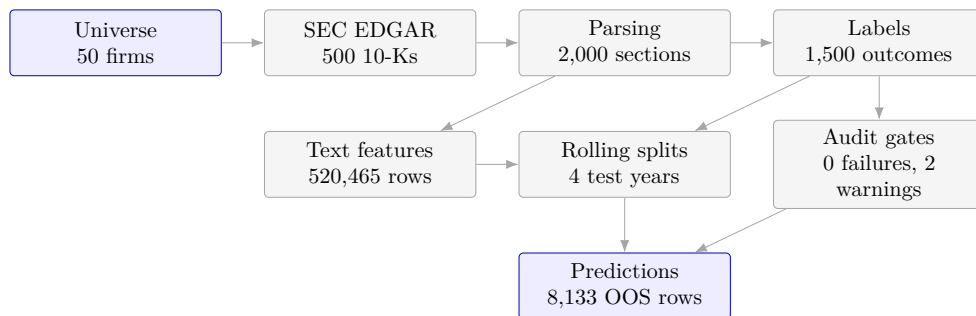


Figure 2: Data construction workflow recorded by the local run artifacts. Each node corresponds to a persisted artifact or audit gate, so the empirical claim can be traced from universe definition to out-of-sample prediction rows.

and zero split records with failure severity. The coverage waterfall reports 744 eligible out-of-sample labels, 603 portfolio-eligible labels, and 120 labels outside the configured validation/test split window. Raw label coverage is therefore 0.496, but eligible out-of-sample coverage, model-expected prediction coverage, primary prediction coverage, and primary portfolio coverage are all 1.000. This distinction matters because the raw denominator includes labels that the rolling design does not intend to test out of sample.

4. Text Parsing and Feature Construction

The text pipeline converts each SEC filing into two linked representations: a full-document text and four economically interpretable 10-K sections. The section scopes are Item 1 Business, Item 1A Risk Factors, Item 3 Legal Proceedings, and Item 7 MD&A. This design keeps the empirical problem close to the structure of the filing. Full-document features summarize the overall disclosure environment, whereas section-level features separate operating description, enumerated risks, litigation exposure, and management discussion. The distinction is useful because a predictive relation may come from total disclosure burden, from risk-specific language, or from a particular part of the filing that is more directly connected to future uncertainty.

4.1. Parsing and section-quality gates

The parser output is treated as an auditable data layer rather than an invisible preprocessing step. The local parsing report records 500 parsed 10-K documents, zero missing raw filings, and 2,000 section records. Each

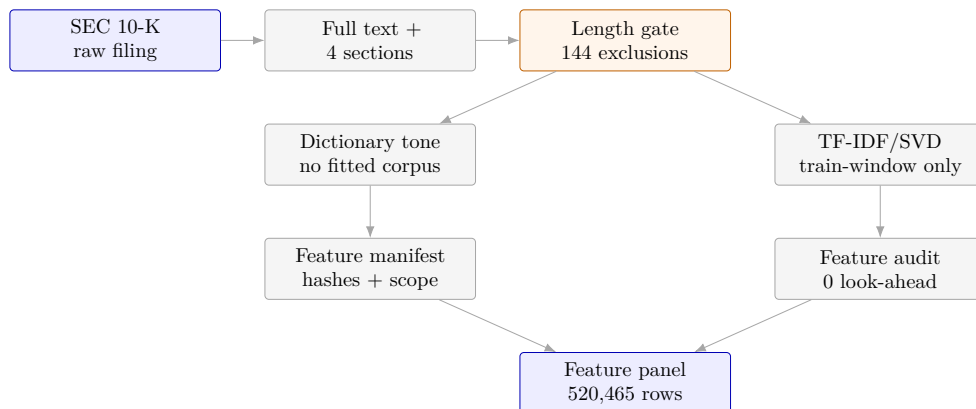


Figure 3: Text parsing and feature-construction logic. Dictionary features require no fitted vocabulary, while TF-IDF/SVD features are refit within each rolling training window and traced through split-specific manifests.

256 parsed section carries a document identifier, ticker, item code, section name,
 257 character count, word count, parser status, quality flag, and feature-usage
 258 policy. These fields allow the paper to distinguish between a document that
 259 is genuinely short and a document whose section boundary may have been
 260 misidentified.

261 The section-length gate is deliberately conservative. Across the 2,000
 262 parsed section records, 1,840 are marked *ok*, 16 receive a warning for fewer
 263 than 500 words, and 144 are marked for manual review below 100 words. The
 264 same artifact records 144 section-level feature exclusions. The exclusion rule
 265 applies to very short Item 1A or Item 7 extractions until parser boundaries are
 266 manually reviewed. The purpose is not to improve model fit mechanically; it is
 267 to prevent a boundary error from becoming a section-level signal. Document-
 268 level features remain available, but a suspiciously short section is not allowed
 269 to enter the section-specific feature space as if it were a faithful disclosure
 270 excerpt.

271 4.2. Dictionary tone features

272 The first feature family uses finance-specific dictionary counts. The con-
 273 figuration enables dictionary-tone extraction using the Loughran–McDonald
 274 dictionary family and risk/uncertainty measures, and the local feature mani-
 275 fest records the Notre Dame SRAF Loughran–McDonald Master Dictionary
 276 as the dictionary source. The manifest version is `LM_1993_2025`, with 3,876
 277 dictionary terms and a license note restricting the raw dictionary material to

278 the private data area. This feature family follows the logic of finance-domain
279 textual analysis: generic word sentiment is not assumed to carry the same
280 meaning in financial reports, where words such as “liability” and “capital”
281 may have domain-specific meanings (Loughran and McDonald, 2011).

282 Dictionary features are constructed for the full document and for each
283 of the four section scopes. The observed feature names include word counts
284 and counts or shares for negative, positive, uncertainty, litigious, strong-
285 modal, weak-modal, and constraining language. Because these measures are
286 dictionary counts, their fit scope is `no_fit_dictionary_counts`. They do
287 not require a corpus-level vocabulary to be estimated from the sample, and
288 they therefore avoid one common source of train-test leakage by construction.
289 The full `features.jsonl` file contains 34,485 dictionary-tone rows and 75
290 unique dictionary feature names.

291 4.3. Train-window TF-IDF/SVD features

292 The second feature family represents higher-dimensional language variation
293 through unigram TF-IDF followed by truncated SVD. The configuration
294 sets a maximum of 10,000 vocabulary terms, unigram tokens, minimum
295 document frequency of 4, maximum document frequency of 0.95, sublinear
296 term frequency, and a token pattern restricted to alphabetic tokens with
297 internal hyphens or apostrophes. For each rolling split and text scope, the
298 TF-IDF vocabulary is fit only on the training window and then applied
299 to validation and test observations. This rule is central to the empirical
300 design. A full-sample vocabulary could silently encode future language
301 distributions, especially when the prediction problem spans 2016–2025 and
302 corporate disclosure vocabularies evolve over time.

303 The local artifacts make this rule observable. The feature manifest con-
304 tains 20 TF-IDF/SVD manifests: four rolling splits crossed with five text
305 scopes. The vocabulary manifest likewise contains 20 train-window vocabulary
306 entries, each with a separate vocabulary hash. The full-document and Item 1
307 vocabularies reach the configured 10,000-term cap in all four splits; Item 1A
308 averages 8,819.5 terms, Item 3 averages 1,659.2 terms, and Item 7 averages
309 9,598 terms. This pattern is substantively plausible because Legal Proceed-
310 ings sections are often short or boilerplate, while Business, Risk Factors, and
311 MD&A sections contain richer language variation.

312 After TF-IDF construction, the pipeline stores 64 SVD components per
313 split and text scope. All 20 TF-IDF/SVD manifests record 64 effective com-
314 ponents. The recorded sum of explained variance across the 64 components

315 ranges from 0.8758 to 0.9509, with an average of 0.9108. These SVD features
316 are not interpreted as topics in the paper. Because SVD components are
317 rotation-dependent compressed predictors, the paper uses them for predictive
318 evaluation rather than coefficient-level economic interpretation. In the full
319 feature panel, TF-IDF/SVD accounts for 484,480 rows and 320 unique com-
320 ponent feature names, corresponding to 64 components for each of the five
321 text scopes.

322 4.4. *Feature manifests and timing discipline*

323 The feature panel contains 520,465 rows: 34,485 dictionary-tone rows,
324 484,480 TF-IDF/SVD rows, and 1,500 metadata rows. Each feature row
325 records the entity identifier, ticker, source document, source chunk, feature
326 family, feature name, feature value, feature version, event time, prediction time,
327 and feature time. The explicit time fields are important because the analysis
328 uses filing events rather than fiscal-year labels alone. The audit report records
329 zero feature rows with `feature_time_utc` after `prediction_time_utc`. It
330 also records zero TF-IDF feature manifests that violate the train-window-only
331 fit-scope requirement.

332 This feature design supports the paper’s main evidentiary posture. It does
333 not claim that the parser perfectly recovers every legal subsection of every
334 filing, nor that TF-IDF/SVD components are stable economic constructs.
335 Instead, it claims that the text variables used in prediction are generated by a
336 reproducible, time-stamped, split-aware pipeline. Dictionary features provide
337 interpretable measures of financial tone and uncertainty; TF-IDF/SVD fea-
338 tures provide a compact representation of broader disclosure language. The
339 v4 evaluation layer uses the same feature provenance to form controlled Ridge
340 ablation variants: combined text, dictionary-only text, TF-IDF/SVD-only
341 text, and industry-plus-text. These variants allow the empirical section to
342 distinguish the pre-specified primary combined-text claim from robustness
343 evidence about which feature block carries the strongest observed ranking
344 signal.

Table 3: Audit-control table: text parsing and feature-construction controls recorded by the local run.

Design element	Local implementation and empirical role
Text scopes	Full document plus Item 1, Item 1A, Item 3, and Item 7. The design separates total disclosure burden from section-specific business, risk, legal, and MD&A language.
Parser coverage	500 parsed filings, zero missing raw filings, and 2,000 section records. This supports a complete document-level text panel for the fixed universe.
Section quality gate	1,840 sections marked <code>ok</code> , 16 warnings below 500 words, and 144 manual-review records below 100 words. The 144 shortest Item 1A/Item 7 records are excluded from section-level features.
Dictionary tone	20 manifests across four rolling splits and five scopes; <code>LM_1993_2025</code> ; 3,876 terms; <code>no_fit_dictionary_counts</code> ; 34,485 feature rows.
TF-IDF/SVD	20 manifests across four rolling splits and five scopes; train-window-only vocabulary; unigram TF-IDF; 10,000 maximum terms; 64 effective SVD components; 484,480 feature rows.
Vocabulary audit trail	20 vocabulary hashes, one for each split-by-scope vocabulary. The long-form vocabulary is not written to a single vocabulary payload; it is summarized by the vocabulary manifest and linked to stored feature-name files.
Leakage checks	Audit report records zero feature records with feature time after prediction time and zero TF-IDF manifests outside train-window-only fit scope.

345 5. Label Construction and Event-Time Alignment

346 The label layer defines the prediction problem at the filing-event level.
347 Each label is attached to a specific 10-K document, not only to a firm-year
348 observation. This distinction is important because the empirical question is
349 whether text available at the filing time predicts subsequent market outcomes.
350 A fiscal-year label would blur the date on which investors could observe the
351 disclosure; the local run instead records the SEC acceptance time, the filing
352 event time, the market-session status, and the forward trading window used
353 to compute each outcome.

354 5.1. Filing event time and prediction time

355 The document manifest provides the temporal anchor for all labels.
356 For each of the 500 filings, the manifest records `available_time_utc`,
357 `acceptance_time_utc`, `event_time_utc`, market open and close times, a
358 trading-day flag, an event-date policy, and a resolved event date. All 500
359 filings in the local run fall on trading days. The event-date policy distribution
360 is 236 intraday filings, 42 pre-open filings, and 222 after-close filings. These
361 fields are retained because the same calendar filing date can have different
362 information timing depending on whether the filing arrives before the open,
363 during the trading session, or after the close.

364 The label builder sets `prediction_time_utc` equal to the filing’s
365 `event_time_utc`. The label schema then enforces a strict ordering rule:
366 prediction time must be before the label start date. In the local labels, the
367 first observation in the return window is therefore the first selected trading
368 day after the event date. For consistency and to avoid partial-day informa-
369 tion assumptions, the current run begins all return windows on the next
370 selected trading day after the resolved filing event date, including pre-open
371 and intraday filings. For example, Microsoft’s 2016 10-K was accepted at
372 2016-07-28 16:12:18 UTC, and its 20-day labels start on 2016-07-29 and end
373 on 2016-08-25. The same event-time convention is used for all three targets.

374 5.2. Forward-window target definitions

375 The current run constructs three targets for each filing: 20-day realized
376 volatility, 5-day cumulative abnormal return, and 20-day cumulative abnormal
377 return. The price panel contains 133,518 rows from December 1, 2015 through
378 April 30, 2026 across 51 tickers, including the SPY benchmark. Returns are
379 close-adjusted log returns in the label construction used by this run. The

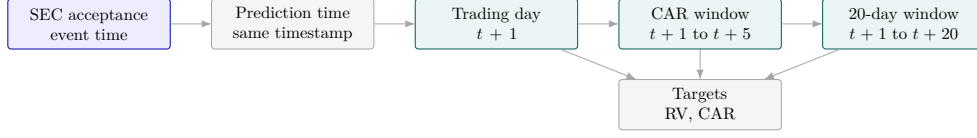


Figure 4: Event-time alignment used for label construction. Prediction time is the SEC filing event time, while realized outcomes begin on the following trading window, preventing same-day or fiscal-year look-ahead from entering the target.

SPY benchmark is used as a transparent market-adjusted return benchmark rather than as a full factor-model abnormal return adjustment. A formal asset-pricing implementation would require richer benchmark models and portfolio tests of the type developed in the factor-model literature (Fama and French, 1993; Carhart, 1997; Fama and French, 2015). The price horizon extends beyond the 2025 filing year so that filings near the end of FY2025 can receive post-filing labels.

Let $r_{i,t+k}$ denote the close-adjusted log return for firm i on the k th trading day after the filing event date, and let $r_{m,t+k}$ denote the corresponding SPY benchmark return. The abnormal-return labels are

$$CAR_{i,1,h} = \sum_{k=1}^h r_{i,t+k} - \sum_{k=1}^h r_{m,t+k}, \quad h \in \{5, 20\}.$$

The realized-volatility label follows the realized-volatility forecasting tradition by summarizing the dispersion of forward daily returns (Andersen et al., 2003; Corsi, 2009). In the current run, it is the sample standard deviation of firm log returns over the 20-day forward window,

$$RV_{i,1,20} = \sqrt{\frac{1}{19} \sum_{k=1}^{20} (r_{i,t+k} - \bar{r}_{i,1,20})^2}.$$

Here $\bar{r}_{i,1,20} = (1/20) \sum_{k=1}^{20} r_{i,t+k}$. This formula follows the sample standard-deviation convention used in the local label builder. The code supports an annualized realized-volatility target, but the configured target in this run is the non-annualized realized-volatility 1–20 target.

5.3. Label completeness and return-quality flags

The label output contains 1,500 rows: 500 labels for each of the three targets. The label-failure file is empty, and all label rows carry the return-quality flag `complete`. The delisting application report records 1,500 labels,

Table 4: Audit-control table: label targets and event-window definitions in the local run.

Target	Window and benchmark	Local implementation
Realized volatility 1–20	Trading days $t + 1$ through $t + 20$; no benchmark	Sample standard deviation of firm close-adjusted log returns; 500 labels; benchmark method recorded as none .
CAR 1–5	Trading days $t + 1$ through $t + 5$; SPY benchmark	Sum of firm log returns minus sum of SPY log returns over the five-day window; 500 labels.
CAR 1–20	Trading days $t + 1$ through $t + 20$; SPY benchmark	Sum of firm log returns minus sum of SPY log returns over the 20-day window; 500 labels.
Return and adjustment fields	Log returns; close-adjusted prices	Label rows record start and end dates, benchmark method, label version, delisting flag, and return-quality flag.

402 zero delisting returns applied, zero labels missing delisting returns, and status
403 **not_applicable**. These facts mean that, within the applied public-data
404 stack used here, the run was able to compute every configured label for every
405 filing.

406 The absence of label failures should not be read as a CRSP-equivalent
407 delisting treatment. The configuration identifies the market data as a mixed
408 FMP/Yahoo public-source stack, and the audit report explicitly warns that
409 market-data-dependent portfolio evidence should not be presented as a
410 CRSP/WRDS-equivalent formal result. Label completeness therefore sup-
411 ports the internal prediction design, but it does not remove the data-boundary
412 limitation for formal asset-pricing claims.

413 5.4. Coverage, eligibility, and split alignment

414 Coverage is measured at two levels. The raw label file has 1,500 rows, but
415 the rolling out-of-sample design only expects labels whose event years fall
416 into the validation or test windows of the four split definitions. The coverage
417 waterfall records 744 eligible out-of-sample labels, 603 portfolio-eligible labels,
418 and 120 labels outside the configured validation/test split window. Eligible
419 out-of-sample coverage is 1.000, primary prediction coverage is 1.000, and

primary portfolio coverage is 1.000; model-expected prediction coverage is 0.9855 because 8,133 of 8,253 expected model-label pairs are populated. Raw label coverage is 0.496 because the denominator includes labels that are not intended to enter out-of-sample prediction under the rolling split design.

The target-level coverage file shows the same distinction. CAR 1–5 has 250 eligible out-of-sample labels, while CAR 1–20 and realized volatility 1–20 each have 247. The split-level coverage file records 294, 297, 294, and 294 eligible out-of-sample labels across the four rolling splits, each with coverage of 1.000. The slight differences across targets and splits arise from the interaction between forward-window length, event timing, and the rolling validation/test calendar. The 120 missing model-prediction pairs belong to the broader model-expected denominator and do not affect the pre-specified primary prediction or primary portfolio coverage.

This event-time label design is central to the interpretation of the later results. The text features are timestamped at the filing event, the labels begin after that event, and the split layer determines which labels are eligible for out-of-sample evaluation. The resulting empirical object is therefore a filing-time prediction panel rather than a static annual panel. That structure is what permits the paper to discuss out-of-sample volatility prediction while remaining cautious about abnormal-return and portfolio interpretations.

6. Rolling Splits and Leakage Controls

The empirical design uses expanding rolling-year splits rather than a single random train-test partition. This choice follows directly from the paper’s forecasting interpretation. A model that predicts post-filing outcomes in 2024 should not learn vocabulary, model hyperparameters, or target behavior from 2024 or 2025 observations. The split layer therefore orders the sample by filing event date, assigns labels to train, validation, and test roles, and records both accepted assignments and purged label windows.

6.1. Expanding rolling-year windows

The local configuration sets the sample window from January 1, 2016 through December 31, 2025, with a minimum of five training years, one validation year, one test year, and a 20-day embargo. This creates four rolling splits. The first trains on filings from 2016–2020, validates on 2021 filings, and tests on 2022 filings. The final split trains on 2016–2023, validates on 2024,

Table 5: Audit-control table: rolling split schedule and local assignment counts.

Test year	Training window	Validation year	Assigned labels	Role counts and coverage
2022	2016–2020	2021	920	626 train, 141 validation, 153 test; eligible OOS coverage 1.000.
2023	2016–2021	2022	1,068	771 train, 147 validation, 150 test; eligible OOS coverage 1.000.
2024	2016–2022	2023	1,218	924 train, 144 validation, 150 test; eligible OOS coverage 1.000.
2025	2016–2023	2024	1,368	1,074 train, 144 validation, 150 test; eligible OOS coverage 1.000.

and tests on 2025. The training window expands over time while validation and test remain one-year forward blocks.

The split-assignment artifact contains 4,574 assignment records. Across all splits, 3,395 records are assigned to training, 576 to validation, and 603 to test. These are assignment counts rather than unique-label counts because the same filing-year label can play different roles in adjacent rolling splits. For example, labels from 2022 are test labels in the first split and validation labels in the second split. This overlap is a feature of rolling evaluation, not a duplication error.

6.2. Embargo and purged label windows

The split builder applies the 20-day embargo to forward-label windows, not merely to event dates. A training label is purged if its label end date reaches the 20-day cutoff before the validation window begins. A validation label is purged if its label end date reaches the 20-day cutoff before the test window begins. This rule matters because the labels are forward-looking outcomes: a filing late in a training year can have a return or volatility window

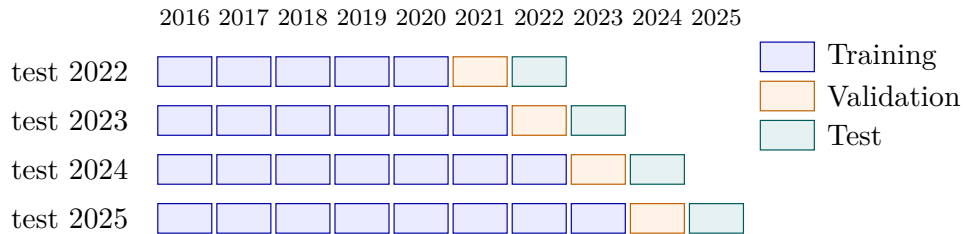


Figure 5: Expanding rolling-year split design. Training years accumulate from 2016, while validation and test advance one year at a time.

that extends close to the start of validation.

The local split-leakage artifact records 46 purged labels and no leakage failures. Of the 46 purges, 22 are training labels whose forward window crosses the validation embargo, and 24 are validation labels whose forward window crosses the test embargo. There are 10 purges in the first split and 12 in each of the remaining three splits. The purges occur only for the two 20-day targets, with 23 records for realized volatility 1–20 and 23 for CAR 1–20. No CAR 1–5 label is purged, which is consistent with its shorter forward window.

6.3. Feature-fitting and model-selection controls

The split layer is only one part of the leakage-control design. The feature layer and model-selection layer must also respect the same temporal ordering. The local feature manifest records 20 TF-IDF/SVD manifests with train-window-only fit scope and 20 dictionary-tone manifests with no fitted corpus. The vocabulary manifest contains split-by-scope vocabulary hashes, so the TF-IDF representation used for a test year is not learned from that test year’s documents. The audit report records zero TF-IDF manifests outside train-window-only fit scope and zero feature rows with feature time after prediction time.

Model tuning follows the same convention. The tuning log contains 84 model-target-split entries after the v4 feature-ablation expansion: historical-mean, industry-mean, combined-text Ridge, Ridge ablation variants, and XGBoost models across targets and rolling splits. The two baseline models use training-window statistics and have no validation search. Ridge, Ridge ablations, and XGBoost use validation rank IC for hyperparameter selection. The audit report records zero tuning logs using a non-validation model-selection metric. Test labels are therefore reserved for out-of-sample evaluation rather than hyperparameter choice.

Table 6: Audit-control table: leakage-control checks recorded by local artifacts.

Control layer	Local evidence and interpretation
Ordered split windows	Split schemas require ordered train, validation, and test windows. The run has four expanding rolling splits and 4,574 assignment records.
Embargo purge	46 labels are purged by the 20-day embargo: 22 train-window purges and 24 validation-window purges. The audit reports zero split records with failure severity.
Event-time label ordering	Label schema requires prediction time before label start date. Label windows begin after the filing event date.
Train-window text fitting	TF-IDF/SVD has 20 train-window-only manifests. Dictionary features use no fitted corpus vocabulary.
Feature timing	Audit reports zero feature records with feature time after prediction time.
Validation-only model selection	Tuning audit reports zero non-validation model-selection metrics; Ridge, Ridge ablation variants, and XGBoost select hyperparameters using validation Rank IC.
Prediction-label alignment	Audit reports zero predictions that fail to map to a valid label identifier.

6.4. Implications for out-of-sample claims

The rolling design defines the scope of the paper’s prediction evidence. Eligible out-of-sample coverage is 1.000 in each of the four splits, but this statement applies only to labels assigned to validation or test roles under the split design. It should not be confused with the raw label coverage ratio, which includes labels outside the configured validation/test windows. The relevant empirical claim is therefore not that every generated label is tested in every split; it is that every label expected by the rolling out-of-sample design receives the required prediction coverage.

This protocol also limits how later results should be interpreted. Rank IC and portfolio diagnostics are computed after a sequence of event-time labels, embargoed rolling splits, train-window-only text features, and validation-only model selection. These controls reduce look-ahead and specification-selection risks, but they do not by themselves convert the applied public-source data stack into a formal asset-pricing database. The correct claim is narrower and more defensible: within the fixed 50-company applied panel, the reported prediction results are generated under a documented time-ordering

514 and leakage-control regime.

515 7. Prediction Design and Specification Governance

516 The prediction layer converts the document-level feature panel into out-
517 of-sample forecasts while preserving the time ordering established in the
518 preceding sections. The v4 local run evaluates historical-mean and industry-
519 mean baselines, combined-text Ridge, three Ridge feature-ablation variants,
520 and XGBoost across the configured targets and rolling splits. This produces
521 84 model-manifest entries. The text models use the recorded split-specific
522 feature representation, while the historical-mean and industry-mean baselines
523 ignore text features and use only training-window target information. Training
524 observation counts range from 208 to 360, validation counts from 46 to 51,
525 and test counts from 50 to 51. These counts reflect the rolling-year design
526 and the label-specific embargo rules rather than arbitrary resampling.

527 The two baselines define what the text models must improve upon. The
528 historical-mean model predicts the training-window target mean and therefore
529 measures the information content of a simple unconditional forecast. The
530 industry-mean model predicts training-window industry means and falls back
531 to the training global mean for industries not observed in the relevant training
532 sample. These baselines are deliberately simple. They do not search over
533 hyperparameters, and their tuning-log entries record `not_applicable` as the
534 validation metric. Their purpose is to keep the empirical comparison anchored
535 to feasible, training-window-only information rather than to a full-sample
536 benchmark.

537 Ridge regression is the primary text model because it is transparent,
538 regularized, and suitable for a feature panel that combines dictionary variables
539 with compressed TF-IDF/SVD components. In every split-target cell, Ridge
540 is fit with standardized features and an intercept. The tuning grid searches
541 five values of the penalty parameter, $\alpha \in \{0.01, 0.1, 1, 10, 100\}$, and selects
542 the value with the highest validation Rank IC. The v4 run adds Ridge
543 feature-ablation variants for dictionary-only text, TF-IDF/SVD-only text, and
544 industry-plus-text. These variants use the same rolling splits, validation-only
545 selection metric, and tuning budget as the primary Ridge model. XGBoost
546 is included as a richer nonlinear comparison rather than as the primary
547 claim. Although the XGBoost parameter grid records candidate domains for
548 estimators, depth, and learning rate, the local search evaluates three predefined

549 shallow-tree configurations in each split-target cell: (50, 2, 0.05), (50, 3, 0.05),
550 and (100, 2, 0.03) for (*n_estimators*, *max_depth*, *learning_rate*).

551 Validation Rank IC is the only model-selection criterion for Ridge, XG-
552 Boost, and the Ridge ablation variants in the recorded tuning logs. The
553 local tuning log contains 84 entries after the v4 feature-ablation expansion.
554 Baseline entries require no hyperparameter search, while text-model entries
555 use `validation_rank_ic`. The audit report records zero tuning logs using a
556 non-validation model-selection metric. This distinction matters because the
557 paper’s target metric is a ranking statistic rather than only a point-forecast
558 loss. A model can be useful for volatility sorting even when its scale is
559 imperfect. Using validation Rank IC therefore aligns model selection with
560 the cross-sectional prediction question while still keeping test-year outcomes
561 unavailable during hyperparameter choice.

562 The trained models generate 8,133 out-of-sample prediction records. Pre-
563 diction coverage is governed by several denominators. Raw label coverage is
564 0.496 because many labels belong to training windows or fall outside config-
565 ured validation/test windows and are not expected to receive out-of-sample
566 predictions. Eligible out-of-sample coverage is 1.000, model-expected pre-
567 diction coverage is 0.9855, and primary-specification coverage is 1.000. The
568 model-prediction failure file contains 120 records in the broader expected-pair
569 denominator, but the pre-specified primary prediction and primary portfolio
570 specifications remain fully covered. These checks make the prediction panel
571 auditable before any claim is made about forecast performance.

572 The specification registry governs how the paper separates primary evi-
573 dence from robustness checks and exploratory comparisons. The local artifact
574 is labeled `specification-registry-v1` and records two primary specifica-
575 tions, 188 robustness specifications, and 404 exploratory specifications, for
576 594 tested specifications in total. The registry artifact itself uses the term
577 `pre_registered`; the paper uses the more conservative term pre-specified be-
578 cause the rules are fixed in the run-level registry before interpretation, not in a
579 public external registry. The first primary specification is the prediction claim:
580 ALL_SPLITS Rank IC for combined-text Ridge on `realized_volatility_`
581 `1_20`, with Fisher-z p-value calculation for Rank IC. The second primary speci-
582 fication is the portfolio diagnostic: ALL_SPLITS monthly common-rebalance
583 sector-neutral equal-weight portfolio Sharpe for the same target and model,
584 with a p-value derived from the Newey-West t-statistic of the out-of-sample
585 return series. This registry design prevents the strongest observed robustness
586 or exploratory model from silently replacing the primary claim.

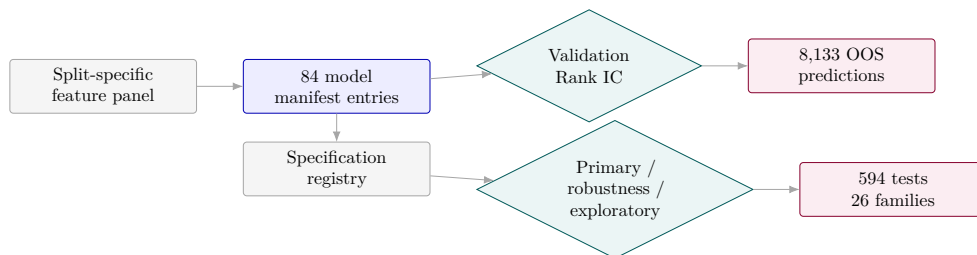


Figure 6: Prediction and specification-governance design. Model selection is performed inside validation windows, while the registry and multiple-testing report determine how prediction, portfolio, robustness, and exploratory claims are labeled.

Multiple-testing disclosure is applied after the registry is constructed. The multiple-testing report records 594 p-values across 26 families and applies Bonferroni, Holm, and Benjamini–Hochberg false-discovery-rate adjustments (Benjamini and Hochberg, 1995). The tested specifications include the primary prediction and portfolio tests, the expanded Ridge feature-ablation comparisons, industry-neutral Rank IC diagnostics, Newey-West statistics, and portfolio-Sharpe families. For the primary realized-volatility Rank IC family, there is one test, so the Bonferroni, Holm, and BH-FDR adjusted p-values all equal the raw p-value of 0.0006677. At the 10 percent FDR threshold, the report records 1 primary discovery, 16 robustness discoveries, and 41 exploratory discoveries. These adjusted results are not used to relabel robustness or exploratory findings as primary findings. They disclose the size and structure of the search space.

This governance structure determines how the next results section should be read. The central question is not which of the 594 specifications has the smallest p-value. The central question is whether the pre-specified primary prediction rule produces out-of-sample evidence under the timing, feature-fitting, and model-selection controls already described. Ridge feature-ablation performance, industry-neutral diagnostics, XGBoost performance, split-level portfolio diagnostics, and alternative return-target specifications remain informative, but they are interpreted as diagnostics around the primary volatility-prediction claim.

Table 7: Prediction models, model-selection rules, and governance role in the local run.

Model class	Local specification	Selection rule	Empirical role
Historical mean	12 model entries; training-window target mean	No hyperparameter search; not_applicable validation metric	Unconditional baseline for target-level predictability.
Industry mean	12 model entries; training-window industry means with global fallback	No hyperparameter search; not_applicable validation metric	Cross-sectional baseline using sector/industry structure available inside the training window.
Ridge combined text	12 model entries; standardized dictionary and TF-IDF/SVD features; intercept; $\alpha \in$ $\{0.01, 0.1, 1, 10, 100\}$	Highest validation Rank IC; ties keep the first grid value	Primary prediction model for the preferred volatility target.
Ridge ablations	36 model entries; dictionary-only, TF-IDF/SVD-only, and industry-plus-text variants	Same Ridge grid and validation Rank IC rule as the primary Ridge model	Registry robustness comparisons; interpreted as diagnostics about feature-block contribution.
XGBoost	12 model entries; shallow tree boosting; three searched parameter combinations per split-target cell	Highest validation Rank IC; ties keep the first grid value	Registry robustness comparison; interpreted as exploratory relative to the Ridge primary specification.

609 8. Empirical Prediction Results

610 The prediction results support a narrow but meaningful claim: 10-K
611 text features contain out-of-sample ranking information for future 20-day
612 realized volatility in the fixed 50-company applied panel. Table 8 reports the
613 selected ALL_SPLITS prediction results used in this section. The evidence
614 is strongest for the realized-volatility target, weaker for the abnormal-return
615 targets, and should not be read as a portfolio-alpha result.

616 8.1. Primary realized-volatility result

617 The pre-specified primary prediction specification is combined-text Ridge
618 on the 20-day realized-volatility target, evaluated by ALL_SPLITS out-of-
619 sample Rank IC. This specification produces a Rank IC of 0.2395 across 197
620 test-role prediction observations from the four test years 2022–2025. The
621 larger coverage counts reported earlier include validation-role labels, unique
622 eligible out-of-sample labels across targets, and model-label pairs across model
623 classes. The raw Fisher-z p-value is 0.00067, and the same value is retained
624 after Bonferroni, Holm, and BH-FDR adjustment within the one-test primary
625 realized-volatility Rank IC family. The Newey-West t-statistic reported for the
626 same prediction metric is 2.3457. For the ALL_SPLITS prediction metrics,
627 the Newey-West statistic is computed on the sequence of split-level Rank IC
628 values rather than on individual prediction errors; with four test splits and
629 configured lag 19, the implementation truncates the effective lag to three.
630 Because this statistic is based on only four split-level observations, it is treated
631 as a diagnostic inference measure rather than as a definitive large-sample
632 test. Because the model, target, split aggregation, and metric are fixed in the
633 run-level specification registry before interpretation, this result is the paper’s
634 primary empirical prediction claim.

635 The result is economically consistent with the disclosure channel described
636 earlier. Risk factors, litigation language, liquidity discussion, operational
637 complexity, and MD&A commentary are more directly related to future
638 uncertainty than to signed abnormal returns. A positive Rank IC therefore
639 indicates that the model tends to assign higher volatility scores to firms that
640 subsequently realize higher 20-day volatility. The evidence does not imply
641 that the scale of the Ridge forecasts is optimal: the primary Ridge RMSE
642 is 0.01932, larger than the industry-mean, TF-IDF/SVD-only Ridge, and
643 XGBoost RMSE values. The primary result is a cross-sectional ranking result,
644 not a calibrated volatility-level forecast.

Table 8: Selected ALL SPLITS prediction results and inference

Model	Target	Role	N	Rank IC	p_{raw}	p_{BH}	NW t	RMSE
Ridge	$RV_{1,20}$	Primary	197	0.2395	0.00067	0.00067	2.3457	0.01932
Ridge TF-IDF/SVD only	$RV_{1,20}$	Rob.	197	0.3668	0.00002	0.00034	5.4055	0.00992
Ridge industry + text	$RV_{1,20}$	Rob.	197	0.3296	0.00008	0.00036	5.1073	0.01076
Industry mean	$RV_{1,20}$	Base	201	0.2924	0.00002	0.00017	4.7084	0.00913
XGBoost	$RV_{1,20}$	Rob.	197	0.2741	0.00009	0.00036	5.6838	0.00836
Ridge	$CAR_{1,20}$	Return	197	-0.0459	0.5205	0.6068	-0.7083	0.12540
Ridge	$CAR_{1,5}$	Return	197	-0.0112	0.8762	0.9856	-0.5586	0.09555

Notes: $RV_{1,20}$ denotes **realized_volatility_1_20**. Rob. denotes registry robustness comparison; these rows are interpreted as exploratory relative to the pre-specified combined-text Ridge primary claim. Base denotes the industry-mean economic baseline. p_{raw} is the Fisher-z p-value for Rank IC. p_{BH} is the Benjamini–Hochberg FDR-adjusted p-value within the corresponding role-target Rank IC family. NW t is the Newey-West t-statistic for the ALL_SPLITS split-level Rank IC sequence; the configured lag is 19 and the effective lag is truncated to $n - 1 = 3$ for the four split-level observations.

645 8.2. Feature ablation and industry-neutral diagnostics

646 The strongest observed model-comparison result in v4 is not XGBoost
647 but Ridge fit on the TF-IDF/SVD block alone. Table 9 reports the feature-
648 ablation diagnostics for the realized-volatility target. The TF-IDF/SVD-only
649 Ridge model produces Rank IC of 0.3668, industry-neutral Rank IC of 0.3416,
650 Newey-West t-statistic of 5.4055, and neutral Newey-West t-statistic of 3.8557.
651 The industry-plus-text Ridge variant is also strong, with raw Rank IC of
652 0.3296 and industry-neutral Rank IC of 0.3251. These rows are informative
653 because they use the same rolling splits, validation-only alpha selection, and
654 tuning budget as the primary Ridge specification.

655 The industry-mean baseline remains strong, with Rank IC of 0.2924 and
656 RMSE of 0.00913. This fact narrows the interpretation of the primary result.
657 The current evidence is predictive, not incremental-causal: it shows that
658 audited text-feature models rank realized volatility out of sample, and that
659 some text blocks retain ranking information after industry neutralization,
660 but it does not yet show that text improves on a full set of non-text volatil-
661 ity controls such as lagged volatility, size, liquidity, analyst coverage, and
662 accounting fundamentals. The industry-neutral diagnostics are therefore a
663 useful v4 improvement, but they remain descriptive rather than a substitute
664 for a research-grade incremental-information design.

Table 9: Feature-ablation and industry-neutral Rank IC diagnostics for realized volatility

Feature set	Estimator	N	Rank IC	Neutral Rank IC	NW t	Neutral NW t	RMSE
Combined text	Ridge	197	0.2395	0.2023	2.3457	1.3704	0.01932
Dictionary only	Ridge	197	0.2244	0.2465	2.9370	5.2660	0.00984
Industry only	Industry mean	201	0.2924	0.0000	4.7084	0.0000	0.00913
Industry + text	Ridge	197	0.3296	0.3251	5.1073	4.3994	0.01076
TF-IDF/SVD only	Ridge	197	0.3668	0.3416	5.4055	3.8557	0.00992
XGBoost combined text	XGBoost	197	0.2741	0.1991	5.6838	2.5256	0.00836

Notes: Industry-neutral Rank IC separately demeans realized targets and predictions within each OOS split-industry group before applying tie-aware rank correlation. It is a descriptive incremental-signal diagnostic, not a causal decomposition. The v4 report records 116 split-industry groups for these diagnostics, including 76 singleton groups.

Table 10: Split-level realized-volatility Rank IC diagnostics

Model	2022	2023	2024	2025	Test obs.
Industry mean	0.3744	0.5609	-0.1322	0.3778	201
Ridge combined text	0.4789	0.4490	-0.0539	0.0841	197
Ridge TF-IDF/SVD only	0.5989	0.5865	0.0328	0.2492	197
XGBoost	0.3898	0.4101	-0.0504	0.3468	197

Notes: Each year column reports test-role Rank IC for `realized_volatility_1_20`. The rows should be read as split-level diagnostics rather than separate primary tests.

665 8.3. Split-level stability

666 Split-level results qualify the ALL_SPLITS result. Table 10 shows that
667 the realized-volatility signal is strongest in the first half of the OOS period
668 and weaker later. The primary ALL_SPLITS result is driven mainly by
669 strong 2022–2023 performance, with weaker evidence in 2024–2025. It is not a
670 single-year artifact, but its stability is limited. This pattern is consistent with
671 an applied pilot: the signal is visible under the audit protocol, yet year-level
672 stability remains an empirical limitation.

673 8.4. Abnormal-return targets

674 The abnormal-return targets do not show comparable text-model evidence.
675 Combined-text Ridge produces Rank IC of -0.0112 for `CAR_1_5` and -0.0459 for
676 `CAR_1_20`. Some feature-ablation and industry-neutral CAR diagnostics are
677 positive, but the raw return-target evidence is weaker and less stable than the
678 realized-volatility evidence. The return-target evidence is therefore not strong

679 enough to support a claim that 10-K text generates robust short-window
680 abnormal-return predictability in the current applied panel.

681 This contrast is substantively important. The weaker CAR results are not
682 a failure of the disclosure premise; they are consistent with the idea that public
683 annual reports may contain information about uncertainty without generating
684 predictable signed abnormal returns. Market prices can incorporate public
685 disclosure quickly, expected returns may reflect risk compensation rather
686 than mispricing, and short-window abnormal returns contain substantial
687 idiosyncratic noise. The empirical pattern therefore supports the paper’s
688 target hierarchy: volatility prediction is the central claim, while abnormal-
689 return outcomes are diagnostic tests of whether the same signal extends to
690 return direction.

691 *8.5. Interpretation of the prediction evidence*

692 The prediction section has three implications for the rest of the paper.
693 First, the primary empirical result survives the timing and leakage controls
694 described in Sections 3–7: document availability, event-time labels, embargoed
695 rolling splits, train-window-only text features, validation-only model selection,
696 prediction coverage checks, and specification governance. Second, model-
697 comparison evidence is more informative in v4 than in the earlier package:
698 the strongest observed volatility-ranking signal comes from TF-IDF/SVD-
699 only Ridge, not from XGBoost, and industry-neutral diagnostics show that
700 some text variation remains after split-industry demeaning. Third, the
701 bootstrap and portfolio layers constrain the claim. A positive volatility-
702 ranking result can be useful for understanding disclosure information, but it
703 does not automatically imply a robust incremental text channel or a tradable
704 long-short return strategy.

705 The portfolio diagnostic therefore tests the economic boundary of the
706 same prediction signal rather than broadening the claim. A larger universe, a
707 CRSP/WRDS-equivalent market-data stack, explicit delisting-return treat-
708 ment, and incremental tests against non-text volatility predictors would
709 determine whether the present volatility signal remains after stronger data
710 and benchmark controls. Within the current public-source run, the ap-
711 propriate conclusion is precise: 10-K text features produce audit-backed
712 out-of-sample evidence for realized-volatility ranking, while return prediction
713 and trading-alpha interpretations remain limited.

Table 11: Primary monthly common-rebalance portfolio diagnostic

Model	Target	Variant	Direction	Sharpe	Raw p	BH p	Interpretation
Ridge	$RV_{1,20}$	SN-EW monthly quintile	long-low-score	-0.8539	0.1147	0.1147	No primary alpha evidence
Best RV robustness family	$RV_{1,20}$	portfolio Sharpe family	registry-defined	–	0.0472	0.6198	No adjusted portfolio discovery

Notes: SN-EW denotes the monthly sector-neutral equal-weight top-bottom quintile portfolio. The Ridge row is the pre-specified primary portfolio diagnostic. The robustness-family row reports the best raw and Benjamini–Hochberg adjusted p-values from the realized-volatility robustness portfolio-Sharpe family; it is included to show why favorable raw portfolio variants are not interpreted as formal alpha evidence.

9. Portfolio Diagnostics and Trading-Alpha Boundary

The portfolio layer asks whether the prediction scores remain economically informative after they are passed through a specified long-short construction rule. It is deliberately framed as a diagnostic layer, not as an asset-pricing test. The primary portfolio specification in the registry uses the Ridge score for `realized_volatility_1_20`, monthly common rebalancing, a top-bottom quintile construction with at least one name per leg, equal weighting, sector-neutral exposure, the `long_low_score` convention for the volatility target, and the configured transaction-cost rule. This design tests whether the volatility-ranking signal can survive a simple trading translation, but it does not supply the research-grade market-data conditions required for a formal alpha claim.

The primary portfolio diagnostic is negative. The Ridge monthly sector-neutral equal-weight volatility portfolio has Sharpe ratio of -0.8539 and registry p-value of 0.1147. Because the primary realized-volatility portfolio-Sharpe family contains one test, the Bonferroni, Holm, and BH-FDR adjusted p-values are also 0.1147. The result therefore fails to support the second primary specification. It also clarifies the difference between prediction and trading interpretation: the same Ridge model that ranks future realized volatility out of sample does not produce a positive primary monthly sector-neutral portfolio diagnostic. The portfolio return sources are the daily price panel and the monthly common-rebalance price panel. Position accounting is recorded through drifted daily positions and monthly-rebalance drifted positions.

Model comparisons do not overturn this boundary. Some split-level or variant portfolio rows have more favorable raw Sharpe ratios, but the multiple-testing report shows that portfolio families do not produce adjusted discoveries. In the realized-volatility robustness portfolio-Sharpe family, the best raw p-

741 value is 0.0472, while the best Benjamini–Hochberg adjusted p-value is 0.6198.
742 These rows are economically useful diagnostics, but they do not convert
743 the prediction result into alpha evidence. XGBoost and the Ridge ablation
744 variants therefore remain robustness comparisons interpreted as exploratory
745 relative to the primary combined-text Ridge specification.

746 The strongest observed event-based or portfolio variants are also insufficient
747 as formal claims. They are not the primary portfolio specification, are often
748 not ALL_SPLITS monthly common-rebalance results, and may be drawn
749 from return targets whose raw prediction evidence is weak in Section 8.
750 Because the v4 specification registry contains broad portfolio families, the
751 correct comparison is not the best raw Sharpe row but the adjusted family-
752 level evidence. That adjusted evidence remains insufficient for a formal
753 trading-alpha claim.

754 The portfolio evidence therefore reinforces the paper’s claim discipline.
755 The run-level factor card sets `formal_result_allowed` to `False`: the
756 run is exploratory, the market data come from a mixed FMP/Yahoo
757 public-source stack, market-cap-at-selection is applied-grade rather than
758 CRSP/Compustat/WRDS-equivalent, the universe is a fixed-company applied
759 panel rather than a survivorship-free research universe, and the audit status
760 is `warn`. These are not pipeline failures. They are the conditions under which
761 the portfolio layer must be read. The current evidence supports an audited
762 volatility-prediction result and a negative primary portfolio diagnostic; it
763 does not support a formal trading-alpha or asset-pricing conclusion.

764 10. Robustness, Audit Diagnostics, and Limitations

765 The robustness evidence in this paper is designed to delimit the empirical
766 claim rather than to search for a stronger one. The v4 run contains 594
767 tested specifications, but the interpretation is organized around the pre-
768 specified combined-text Ridge realized-volatility prediction result, the registry
769 robustness comparisons, and the portfolio boundary tests. This section
770 therefore reads the audit layer as part of the evidence: a result is more
771 useful when the reader can see not only where it succeeds, but also where the
772 data, timing protocol, inference design, and market-data environment prevent
773 stronger claims.

774 *10.1. Coverage and audit diagnostics*

775 Table 12 summarizes the main robustness and audit diagnostics. The
776 most important coverage distinction is between raw label coverage and el-
777 igible out-of-sample coverage. The raw coverage rate is 0.496 because the
778 denominator includes labels outside the configured validation/test split win-
779 dow. Conditional on the labels that are eligible for out-of-sample evaluation,
780 coverage is complete: 744 eligible out-of-sample labels are predicted, and both
781 the primary prediction and primary portfolio specifications have coverage of
782 1.000. The broader model-expected prediction denominator is less than com-
783 plete: 8,133 of 8,253 expected model-label pairs are populated, for coverage of
784 0.9855. This distinction matters because a low raw denominator or a broader
785 model-pair denominator can look like model attrition unless each is tied back
786 to the split design and registry role.

787 The leakage diagnostics are central to the robustness interpretation. The
788 audit report records zero feature rows with feature time after prediction time,
789 zero TF-IDF feature manifests outside the train-window-only fit scope, and
790 zero tuning logs using a non-validation model-selection metric. The split
791 audit records 46 embargo purges and zero failure-severity split records. These
792 purges are economically meaningful because they apply to forward label
793 windows, not only filing dates. A late training-window filing whose 20-day
794 label overlaps the validation or test period is therefore removed rather than
795 allowed to carry future return information into model development. The
796 paper treats the purged observations as the price of a credible timing protocol.

797 *10.2. Bootstrap confidence intervals*

798 The v4 package adds bootstrap confidence intervals for the primary Rank
799 IC estimand. The bootstrap report uses 2,000 deterministic resamples with
800 random seed 42 and reports three complementary clustering schemes: split
801 bootstrap, event-date bootstrap, and ticker-cluster bootstrap. Table 13
802 summarizes the results. For the raw primary ALL_SPLITS Rank IC, the
803 split-bootstrap interval includes zero, while the event-date and ticker-cluster
804 intervals are positive. For the industry-neutral primary Rank IC, all three
805 bootstrap intervals include zero.

806 The bootstrap evidence strengthens the paper’s caution. The primary
807 raw ranking result is positive, but the most conservative split-level bootstrap
808 interval includes zero. The primary industry-neutral point estimate is also
809 positive, but all three industry-neutral bootstrap intervals include zero. The
810 appropriate inference is therefore not that the primary result has become

Table 12: Robustness and audit diagnostics

Layer	Local artifact value	Interpretation
Run-level audit	0 failures; 2 warnings	Internally complete; warnings are boundary disclosures.
Coverage	Raw 0.496; eligible OOS 744/744; model expected 8,133/8,253; primary prediction 1.000; primary portfolio 1.000	Coverage is complete for eligible labels and primary specs, but not for every broader model-label pair.
Split controls	46 embargo-purged records; 0 failure-severity records	Purging is a leakage-control cost, not a failed check.
Feature and tuning leakage	0 late feature records; 0 out-of-scope TF-IDF manifests; 0 non-validation tuning logs	Feature fitting and model selection respect the time-ordering protocol.
Parsing quality	2,000 section records; 1,840 ok; 16 warnings; 144 manual-review flags	Short uncertain section extractions are excluded from section-level features.
Multiple testing	594 p-values; 26 families; 2 primary, 188 robustness, 404 exploratory specifications	Primary results are separated from robustness and exploratory families.
Formal-result boundary	Formal-result flag is false; mixed FMP/Yahoo stack; fixed applied panel	The run supports exploratory prediction evidence, not formal trading alpha.

Table 13: Bootstrap confidence intervals for primary Rank IC estimands

Estimand	Bootstrap method	Point	95% CI	SE	Zero p	Clusters
Raw Rank IC	Split	0.2395	[-0.0050, 0.4841]	0.1069	0.111	4
Raw Rank IC	Event date	0.2395	[0.0719, 0.3743]	0.0759	0.005	104
Raw Rank IC	Ticker cluster	0.2395	[0.1091, 0.3522]	0.0620	0.001	49
Industry-neutral Rank IC	Split	0.2023	[-0.1157, 0.5202]	0.1440	0.117	4
Industry-neutral Rank IC	Event date	0.2023	[-0.1546, 0.4273]	0.1493	0.366	104
Industry-neutral Rank IC	Ticker cluster	0.2023	[-0.1787, 0.4181]	0.1536	0.357	49

Notes: Bootstrap resampling preserves each selected cluster as a block. The split bootstrap has only four clusters, so it is the most conservative and least statistically stable interval. Industry-neutral Rank IC is computed after demeaning targets and predictions within OOS split-industry groups.

a formal incremental text effect. It is that the realized-volatility ranking evidence survives the registry’s primary prediction protocol, while bootstrap and industry-neutral diagnostics require a conservative interpretation.

10.3. Multiple testing and portfolio boundary

The multiple-testing layer supports the same restrained reading. The tested specification file contains 594 p-values across 26 families, including 2 primary, 188 robustness, and 404 exploratory specifications. Within the primary realized-volatility Rank IC family, there is only one test, so the raw, Bonferroni, Holm, and Benjamini-Hochberg adjusted p-values for the Ridge primary prediction result are identical at 0.0006677. At the 10 percent FDR threshold, the report records 1 primary discovery, 16 robustness discoveries, and 41 exploratory discoveries. These adjusted results disclose the size and structure of the search space; they do not relabel robustness or exploratory findings as primary findings.

The portfolio families are broader and do not convert the prediction result into a trading-alpha claim. For realized-volatility portfolio Sharpe tests, the robustness family has 39 tests, a best raw p-value of 0.0472, and a best Benjamini-Hochberg adjusted p-value of 0.6198. The primary portfolio Sharpe is negative, with Sharpe ratio -0.8539 and raw p-value 0.1147. The appropriate inference is therefore asymmetric: the volatility-ranking result survives the primary prediction protocol, while portfolio evidence remains diagnostic.

Several robustness checks also identify limits rather than confirmations. The split-level results in Table 10 show that the Ridge realized-volatility Rank

835 IC is stronger in 2022 and 2023 than in 2024 and 2025. This pattern does not
836 eliminate the ALL_SPLITS result, but it reduces confidence that the signal is
837 stable across market regimes. The Newey-West statistics should also be read
838 cautiously. For the prediction results, the reported statistic is computed on
839 the four split-level Rank IC observations, with the configured lag truncated
840 by the number of available split observations. With only four test splits, this
841 statistic is useful as a diagnostic dependence adjustment rather than as a
842 definitive large-sample inference device.

843 *10.4. Data and parser limitations*

844 Parsing quality is another explicit limitation. The parser produces 2,000
845 section records from 500 filings, and the full-document layer is available for
846 all documents. However, the section-length quality report flags 144 section
847 records for manual review because the extracted section is shorter than 100
848 words, and these records are excluded from section-level features under the
849 local usage policy. This decision prevents boundary errors in short Item 1A
850 or Item 7 extractions from becoming spurious section-level predictors. It also
851 means that section-specific evidence should be read as conditional on the
852 parser’s inclusion rules, not as a complete measurement of every disclosure
853 section in the raw filings.

854 The main data limitation is not a hidden defect in the code path,
855 but the applied nature of the run. The universe is a fixed 50-company
856 panel selected for an auditable public-source pilot, not a survivorship-free
857 CRSP/Compustat/WRDS universe. Market data come from a mixed
858 FMP/Yahoo adjusted-price stack, market capitalization at selection is
859 applied-grade, and the return labels use transparent market-adjusted SPY
860 benchmarks rather than a full factor-model abnormal-return system. The
861 delisting report records no applied or missing delisting returns in this run,
862 and no portfolio positions are affected by delisting, but the absence of
863 a CRSP-style delisting-return integration remains a boundary for formal
864 asset-pricing interpretation. The industry-neutral Rank IC diagnostics are
865 also sparse: the v4 artifact records 116 split-industry groups, including 76
866 singleton groups.

867 These limitations are not incidental qualifications; they define the eviden-
868 tiary boundary of the current run. Until a larger research-grade replication
869 is completed, the paper’s strongest claim remains deliberately narrow: an
870 audited 10-K text pipeline produces out-of-sample realized-volatility rank-
871 ing evidence, while return prediction, portfolio performance, and formal

872 asset-pricing interpretations remain limited by design and by data boundary.

873 11. Conclusion and Future Work

874 This paper asks whether annual 10-K disclosures contain predictive infor-
875 mation for subsequent market outcomes when the text pipeline is constrained
876 by document-level timing, rolling out-of-sample splits, train-window-only
877 feature fitting, validation-only model selection, and explicit multiple-testing
878 disclosure. The answer is qualified but informative. In the current 50-company
879 public-source v4 run, the strongest supported result is not a return-alpha
880 finding. It is an audited realized-volatility ranking result: the pre-specified
881 combined-text Ridge model for `realized_volatility_1_20` produces pos-
882 itive ALL_SPLITS Rank IC evidence under the paper’s event-time and
883 leakage-control rules. This finding is economically plausible because annual
884 disclosures describe operating risk, business complexity, litigation exposure,
885 liquidity constraints, and managerial discussion, all of which may map more
886 directly into future uncertainty than into short-window abnormal returns.

887 The v4 package sharpens this conclusion by separating the primary claim
888 from the strongest observed robustness evidence. The primary combined-text
889 Ridge result has Rank IC of 0.2395 and raw p-value of 0.00067. The strongest
890 observed volatility-ranking result is TF-IDF/SVD-only Ridge, with Rank IC
891 of 0.3668 and industry-neutral Rank IC of 0.3416. This does not replace the
892 primary claim, but it identifies where the text signal is most visible in the
893 current run. At the same time, the bootstrap evidence requires caution: the
894 split-bootstrap confidence interval for the primary raw Rank IC includes zero,
895 and all primary industry-neutral bootstrap intervals include zero. The result
896 should therefore be read as audited exploratory prediction evidence, not as a
897 completed incremental-information or asset-pricing claim.

898 The paper’s main contribution is methodological as much as predictive.
899 It demonstrates a reproducible workflow in which the empirical claim can be
900 traced from SEC filing retrieval through parsing, feature construction, label
901 alignment, rolling splits, model selection, prediction records, evaluation met-
902 rics, feature-ablation diagnostics, portfolio diagnostics, bootstrap inference,
903 multiple-testing reports, and audit reports. The document manifest fixes
904 filing availability and event-time resolution; the parser separates full-document
905 and section-level text while excluding short uncertain sections from section-
906 level features; the feature layer combines finance-specific dictionary measures
907 with train-window-only TF-IDF/SVD representations; and the specification

908 registry separates primary, robustness, and exploratory tests. These design
909 choices do not make the sample large or the market-data stack research-grade,
910 but they make the evidentiary boundary observable.

911 The empirical boundary is equally important. Return-target evidence is
912 weaker for `CAR_1_5` and `CAR_1_20`, and the portfolio layer does not support
913 a formal trading-alpha interpretation. The primary portfolio diagnostic is
914 negative, with Sharpe ratio -0.8539 and p-value 0.1147. The factor card
915 explicitly blocks formal result treatment because the run is exploratory, uses
916 a mixed FMP/Yahoo public-source market-data stack, relies on applied-grade
917 market-cap estimates, and studies a fixed-company applied panel rather than
918 a survivorship-free CRSP/WRDS universe. The results should therefore be
919 read as audited exploratory prediction evidence, not as an asset-pricing factor
920 claim.

921 Future work should convert this applied pilot into a research-grade repli-
922 cation design. The first extension is a broader and survivorship-controlled
923 universe built from CRSP/Compustat or WRDS identifiers, with explicit
924 delisting-return treatment and licensed market-cap histories. The second is
925 an incremental-information design that tests whether 10-K text improves
926 realized-volatility forecasts after controlling for industry, size, past volatility,
927 turnover, analyst coverage, accounting fundamentals, liquidity, and other
928 standard non-text predictors. The third is a stability design: the signal should
929 be re-estimated across a larger set of filing years and market regimes to deter-
930 mine whether the stronger 2022–2023 evidence persists outside the present
931 50-company panel. A final extension would compare dictionary, TF-IDF/SVD,
932 embedding, and large-language-model features under the same locked timing
933 and specification-governance rules, so that richer representations are evaluated
934 by their incremental out-of-sample contribution rather than by post-hoc fit
935 (Lopez-Lira and Tang, 2023). Portfolio extensions should add deflated Sharpe,
936 CPCV/PBO diagnostics, and explicit research-grade transaction-cost and
937 delisting-return treatment before any trading-alpha claim is considered.

938 Taken together, the evidence supports a narrow conclusion. SEC 10-K
939 text can be turned into an auditable out-of-sample signal for future realized-
940 volatility ranking in this applied panel, but the current run does not establish
941 robust abnormal-return predictability, a robust primary industry-neutral
942 incremental effect, or tradable alpha. That distinction is the central discipline
943 of the paper. The contribution is not simply that a text model produces a
944 positive statistic; it is that the statistic is embedded in a transparent workflow
945 that records what was known when, which transformations were fit on which

946 data, which specifications were primary, and where the data boundary stops
947 the claim. A larger research-grade replication can now test whether the same
948 auditable design yields a formal empirical result.

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